Constrained Deep Reinforcement Learning with Lagrangian Theory

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Abstract

Reinforcement learning (RL) agents need to explore their environments in order to learn optimal policies by trial and error. While it is currently typical to train RL agents mostly or entirely in simulation, where safety concerns are minimal. However, it is very difficult to accurately describe complex real-world environments using simulations. Therefore, efforts to train and validate in real-world environments are continuously ongoing. Thus, research on constrained RL has emerged to ensure the stability of agents. In this paper, we propose an algorithm that estimates the Lagrangian multiplier at each step to impose appropriate constraints on each state, ensuring stability.

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1 Introduction

1.1 Background

1.2 Research Objectives

The main objectives of this research are as follows:

- Mathematical stability analysis of the controller and estimator with deep neural networks using the contraction theory.
- Development of the controller and estimator with deep neural networks using the contraction theory.

2 Notations and Preliminaries

An overview of Constrained Markov Decision Process (CMDP) can be found in this book [1].

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2.1 Constrained Markov Decision Process

The constrained Markov decision process (CMDP), $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, r, c \rangle$, is to maximize the expected return of rewards while satisfying the constraint on the expected return of costs:

$$\max_{\pi} J(\pi) = \mathbb{E}_{\tau \sim \pi} \left\{ \sum_{t=0}^{\infty} \gamma^t r_t \right\} \quad \text{s.t. } C(\pi) = \mathbb{E}_{\tau \sim \pi} \left\{ \sum_{t=0}^{\infty} \gamma_c^t c_t \leq d \right\}$$

where $\mathcal{S}$ and $\mathcal{A}$ are state and action space, π is the stationary policy[1]in stationary policy set Π, $\mathcal{P}$ is the environment transition probability. $r, c : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ are the reward and cost functions, respectively, and γ, γ_C are their discounted factors, $\tau = \{s_0, a_0, s_1, a_1, \cdots\}$ is the trajectory under policy π starting from an initial state distribution ρ. The feasible policy set is defined as follows, where $d_i \in \mathbb{R}$:

$$\Pi_C = \{\pi \in \Pi \mid \forall i, \mathcal{J}_{C_i}(\pi) \leq d_i\}$$

In the definition of a CMDP, constraints are typically formulated in a cumulative or probabilistic form. However, in real-world applications, many critical constraints must be satisfied instantaneously and deterministically.

2.2 State-wise Constrained Markov Decision Process

In this research, we are specifically interested in a special type of CMDP where the safety specification is to persistently satisfy a hard cost constraint at every step, which is called State-wise Constrained Markov Decision Process (SCMDP). SCMDP requires the cost for every state-action transition satisfies a hard constraint. The set of feasible stationary policies for SCMDP is defined as follows:

$$\bar{\Pi}_C = \{\pi \in \Pi \mid \forall (s_t, a_t, s_{t+1}) \sim \tau, \ \forall i, \ C_i(s_t, a_t, s_{t+1}) \leq w_i\}$$

where $\tau \sim \pi$ and $w_i \in \mathbb{R}$. Then, the objective for SCMDP is to find a feasible stationary policy from $\bar{\Pi}_C$ that maximizes the performance measure.

$$\max_{\theta} \mathcal{J}(\pi_\theta), \quad \text{s.t. } \pi_\theta \in \bar{\Pi}_C$$

The existing research on state-wise safe RL can be divided into two categories based on whether safety is ensured during training.

- **State-wise safety after convergence.** When a policy converges, it ensures state-wise safety. But the policy may be unsafe during training.
- **State-wise safety during training.** The policy derived from this class of methods always ensures constraint satisfaction, even during training.

In addition, each category can be further classified into two approaches: hierarchical agents and end-to-end agents. In the following sections, we will explore these categories through the lens of these two approaches.

2.2.1 State-wise Safety After Convergence

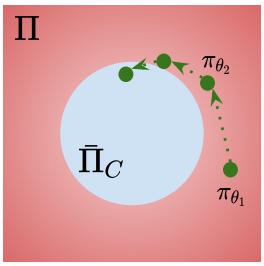

Figure 1: State-wise safety after convergence (reproduced from [2])

State-wise safety after convergence refers to methods that have possible infeasible policies during training and feasible converged stationary policies in the end. As shown in Figure 1, this class of methods uses soft penalties (typically called safe critic $Q_C(s, a)$) to guide the training of policies. The safety critic $Q_C(s, a)$ can be provided by humans or learned from the environment. This class of methods does not pose hard constraints on the exploration space, therefore the policy learning can be made efficient. Nevertheless, the policy may be unsafe during training and the convergence to Π_C.

[1]

1. In CMDP, stationary policies are used to ensure convergence and for mathematical convenience.
2. A stationary policy selects actions based only on the current state, without changing over time, and chooses actions according to the same probability distribution for the same state. Please refer to the book [1] for more details.

2.2.2 State-wise Safety During Training

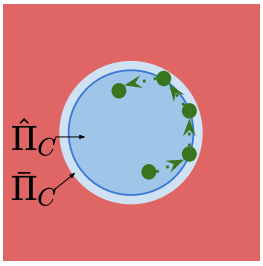

Figure 2: State-wise safety during training by constraining policies in a subset of $\bar{\Pi}_C$ (reproduced from [2])

We denote the policy sequence during training as $\pi_{\theta_1}, \pi_{\theta_2}, \cdots, \pi_{\theta_n}$, where θ_i represents the policy parameters at the i-th optimization iteration, and n is the total number of iterations. State-wise safety during training requires that all intermediate policies satisfy the constraint, i.e., $\pi_{\theta_i} \in \bar{\Pi}_C$ for all $i \leq n$. Becuause it requires all intermediate policies during training to be feasible. As shown in Figure 2, the first class defines a subset of $\bar{\Pi}_C : \hat{\Pi}_C$ (typically by control barrier functions). Whenever the policy goes out of the subset, it will be projected back. Therefore the policy is always guaranteed to be safe during training. However, $\hat{\Pi}_C$ requires a significant amount of prior knowledge about the environment, which may be lacking in many cases. And in general, these methods are conservative than after-convergence methods.

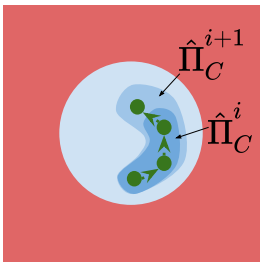

Figure 3: State-wise safety during training by progressive safe exploration (reproduced from [2])

Second class of methods progressively explores the policy space from a safe initial policy as shown in Figure 3. This class of methods maintains an up-to-date subset of $\bar{\Pi}_C : \hat{\Pi}_C$ by exploring and gathering information. The policy updates in $\hat{\Pi}_C^{i+1}$ only if there is enough information to confirm that the $\hat{\Pi}_C^{i+1} \subseteq \bar{\Pi}_C$. This class of methods requires less prior knowledge than the first class but is more conservative in exploration.

3 Related Work

3.1 Constrained Markov Decision Process

3.1.1 Feasible Actor-Critic

The Feasible Actor-Critic (FAC) [3] considers state-wise safety constraints by estimating the Lagrangian multiplier through a multi-layer perceptron (MLP). In this paper, v_C^π, called to as the cost value function, is defined as the expected cumulative cost obtained by following the policy π from an initial state s. Given a policy π, a state s is considered feasible if it satisfies the following inequality.

$$v_C^\pi(s) = \mathbb{E}_{\tau\sim\pi}\left\{\sum_{t=0}^{\infty} \gamma_c^t c_t | s_0 = s\right\} \leq d$$

The feasible region under the policy π is defined as follows.

$$S_F^\pi = \{s | v_C^\pi(s) \leq d\}$$

The set of initial states is denoted by $\mathcal{I}$. Since $\mathcal{I}$ may contain infeasible states, the author considers the intersection of $\mathcal{I}$ and S_F, defined as $\mathcal{I}_F = \mathcal{I} \cap S_F$, which represents the set of feasible initial states. The state-wise safety constrained is defined to guarantee every state in the feasible initial state set to be safe:

$$v_C^\pi(s) \leq d, \quad \forall s \in \mathcal{I}_F$$

As a result, the set of feasible policies[2] under state-wise safety constraints is given by:

$$\Pi_C = \{\pi \mid v_C^\pi(s) \leq d, \forall s \in \mathcal{I}_F\}$$

Accordingly, the optimization problem[3] is formulated as follows to ensure that state-wise safety constraints are satisfied.

$$\max_\pi J(\pi) = \mathbb{E}_{\tau\sim\pi}\left\{\sum_{t=0}^{\infty}\gamma^t r_t\right\} \quad \text{s.t. } v_C^\pi(s) \leq d, \forall s \in \mathcal{I}_F$$

4 Conclusion

References

[1] E. Altman, *Constrained Markov decision processes.* Routledge, 2021.

[2] W. Zhao, T. He, R. Chen, T. Wei, and C. Liu, "State-wise safe reinforcement learning: A survey," 2023.

[3] H. Ma, Y. Guan, S. E. Li, X. Zhang, S. Zheng, and J. Chen, "Feasible actor-critic: Constrained reinforcement learning for ensuring statewise safety," 2021.

[2] This is different from the feasible policy set in a standard Constrained Markov Decision Process (CMDP). Feasible policy set in CMDP: $\Pi_C\{\pi \mid C(\pi) \leq d\}$

[3] This is different from the optimization problem in a standard CMDP. CMDP is to maximize the expected return of rewards while satisfying the constraint on the expected return of costs: $\max_\pi J(\pi) = \mathbb{E}_{\tau\sim\pi}\left\{\sum_{t=0}^{\infty}\gamma^t r_t\right\} \quad \text{s.t. } C(\pi) = \mathbb{E}_{\tau\sim\pi}\left\{\sum_{t=0}^{\infty}\gamma_c^t c_t \leq d\right\}$